

Measuring Planck's constant using the Photoelectric effect

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Abstract

When light of a certain frequency is incident on a metal surface, that surface will emit electrons in response. We call these emitted electrons 'photoelectrons' and this phenomenon the 'photoelectric effect'. In this experiment, we use this phenomenon to measure the value of Planck's constant, h , which is one of the most important constants in modern physics. A Hg lamp has been used as the light source to shine light on the surface of a photodiode and a digital multimeter to measure the stopping potential of the electrons on the diode. Using this method, the value of h is found to be $(5.5 \pm 0.12) \cdot 10^{-34}$ Js which is in relatively close agreement with the current accepted value of $6.63 \cdot 10^{-34}$ Js. [5]

Introduction

Max Planck first proposed the idea of this constant h in 1900 as he was trying to find a mathematical expression that predicted the observed spectral distribution of black-body radiation. He succeeded and the expression now bears his name (Planck's law). In 1901, he calculated the value of h to be $6.55 \cdot 10^{-27}$ Js [1] in a paper published in the 'Annalen der Physik' physics journal. Today, the accepted value of h is $6.63 \cdot 10^{-34}$ Js. For his discovery, Planck recieved the 1918 Nobel Prize in Physics.

In this paper, he specifically suggested to look at the vibrational energy of N oscillators as a discrete quantity which is made up of a finite number of 'energy elements', ε . [1] In this case, the oscillators he considered were molecules. When this new approach was applied to Wien's displacement law, it was shown that ε must be proportional to the frequency f of the oscillator. We now call this the Planck-Einstein relation: $\varepsilon = hf$. [2]

We use this relation when describing the observations of the photoelectric effect. The first mathematical description of this phenomenon came from Albert Einstein in 1905 [3] which earned him the Nobel Prize in Physics in 1921. He also took a similar route to what Planck had adopted in 1900 as he described the effect in terms of 'light quanta' (photons). Additionally, he came up with the concept of a 'work function' ϕ which is a threshold energy below which a photon will not cause a photoelectron to be emitted. [3] Any 'excess' energy would contribute to the electron's kinetic energy K_e . [3]

$$hf = \phi + K_e \quad (1)$$

In this experiment, we used this equation to measure values of both Planck's constant and the work function of the photodiode.

Theory

As described before, the energy of one quantum of electromagnetic radiation ε (in this case light) is directly proportional to its frequency f with h being the constant of proportionality. [2] The energy of one electron can also be found by the equation

$$K_e = eV \quad (2)$$

where e is the electron charge and V is the voltage across the electron [4]. If an electron is repelled from a surface, we can measure the voltage on the surface and use equation (2) to calculate the kinetic energy of the electron. This is called the 'stopping potential' V_s . [4] Using equations (1) and (2), we can construct the following equation.

$$hf = \phi + V_s e \quad (3)$$

This is now much more useful as it includes two measurable quantities: frequency and voltage. One important thing to note is that when the kinetic energy of the photoelectron is 0eV, then the energy of the photon is exactly equal to the work function of the material. So, if we consider the equation $hf = V_s e$ and rearrange it so that it's in the form

$$\frac{V_s}{f} = \frac{h}{e} \quad (4)$$

then we can measure V_s for different frequencies and obtain a straight-line graph with gradient h/e . The y-intercept for such a graph will be the work function ϕ by virtue of the property described above. So, we used this technique to find h , by multiplying the gradient of the graph produced by equation (4) by the electron charge, and ϕ by simply reading off the y-intercept.

Method and Apparatus

The method for this experiment can be separated into 3 parts: (a) setting up the apparatus, (b) investigating how light intensity affects V_s (although this is optional and is only to illustrate that intensity has no effect on the photoelectrons' energies) and (c) measuring stopping potentials for different spectral lines to find h .

The light source we used was a mercury lamp. We also used a diffraction grating to break up the light emitted into its different spectral lines. This is really important as this way we had multiple frequencies of light to work with and so we could create a straight-line graph of V_s against f .

(a) setting up the apparatus

So, we began by setting up the apparatus. The figure below shows the equipment needed.

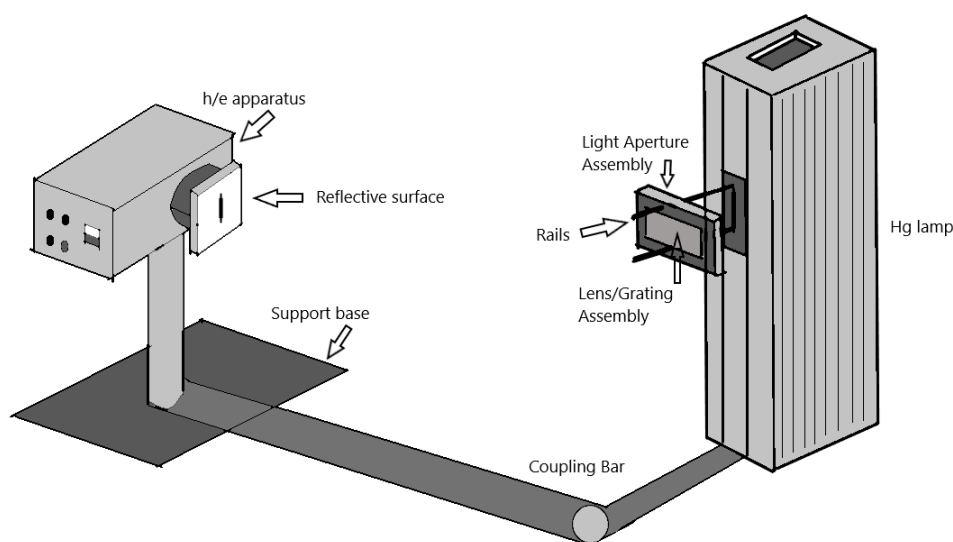


Figure 1: *diagram of mercury lamp and h/e apparatus joined by a rotatable coupling bar.*

In this experiment we also needed a stopwatch, a density wedge filter and a yellow light filter. The density filter allowed us to only let a certain intensity of light through and it's divided into 20%, 40%, 60% and 80% strips which allow in these respective intensities of light. The two filters were placed (magnetically) on the reflective surface in part (b) and the stopwatch was used to measure the time it took for the voltage across the h/e apparatus to arrive to a stable voltage (V_s).

Before starting the experiment, we needed to switch on the Hg lamp and allow it to warm up for at least 30 minutes. This was important for part (b) as we needed the intensity of the lamp to reach a certain minimum so that we could detect any effect it may have had on the photons' energies. (*Note: turning the lamp on and off depending on when it's used wears it more than if it's continuously turned on, so we kept it on throughout the experiment and only turned it off at the end.*)

Once the lamp was warm enough, we rotated the h/e apparatus on the coupling bar so that the spectrum of the light source was incident on the white reflective surface. We then focused the light on this surface by sliding the lens on the rails of the Hg lamp. Now, we made sure that there were 5 distinct visible colours of the spectrum. On the next page is a table of the colours and their respective wavelengths.

Colour	Wavelength(nm)
Yellow	578
Green	546
Blue	436
Violet	405
Ultra-Violet	365

Table 1: *table of Hg spectral line colours and wavelengths.*

We did this by turning the h/e apparatus on the coupling bar counterclockwise since the clearer lines appeared on that side. Finally, we removed the light shield so that the photodiode was revealed and we centred the green spectral line so that it was incident on the photodiode. We made sure that the green line was clearly visible and started part (b).

(b) investigating relationship between V_s and light intensity

We started by placing the yellow filter on the reflective surface and rotating the h/e apparatus so that the first-order yellow light was incident on the photodiode. Then, we placed the density filter on the yellow filter such that the yellow spectral line passed through the 20% intensity section.

Next we needed to connect a digital multimeter as a voltmeter across the h/e apparatus so that we could measure V_s . Once this was done, we fully zeroed the apparatus by holding the discharge button on its side until the voltmeter reading was 0V. Letting go of the discharge button, we timed how long it took for the voltage to stabilise (for the capacitor in the h/e apparatus to charge). We repeated this 5 times for the 20% strip, taking an average for the time, and once for all other strips. Lastly, we repeated all of this with the green spectral line.

(c) measuring Planck's constant, h

The main idea for this last part (as discussed in the Theory section) is to measure V_s for all frequencies of the spectrum and then producing a straight-line graph. The frequencies can be calculated using the equation $c = f\lambda$ where c is the speed of light as the wavelengths of the spectral lines are well known.

However, here we didn't need the density wedge since we were only considering changing frequencies and needed to let all light through. So, first we removed the density wedge and readjusted the h/e apparatus such that the first-order yellow light fell on the photodiode. Then, just as in part (b), we zeroed the apparatus and waited for the voltage to stabilise on the stopping potential (not measuring time). We then repeated this for all 5 colours in the spectrum and plotted a graph of V_s against f .

Experimental Results

We have found that the time taken for voltage to stabilise was inversely proportional to the intensity of light. Increasing the intensity of light means more photons are incident on the photodiode. This in turn means that more energy per unit area is transferred to the surface of the material, hence more photoelectrons are released per unit time. If more photoelectrons are released, then the voltage will increase faster and so the time it takes for the voltage to 'build up' to V_s decreases. Figures 2 and 3 on the next page show this for yellow and green light respectively.

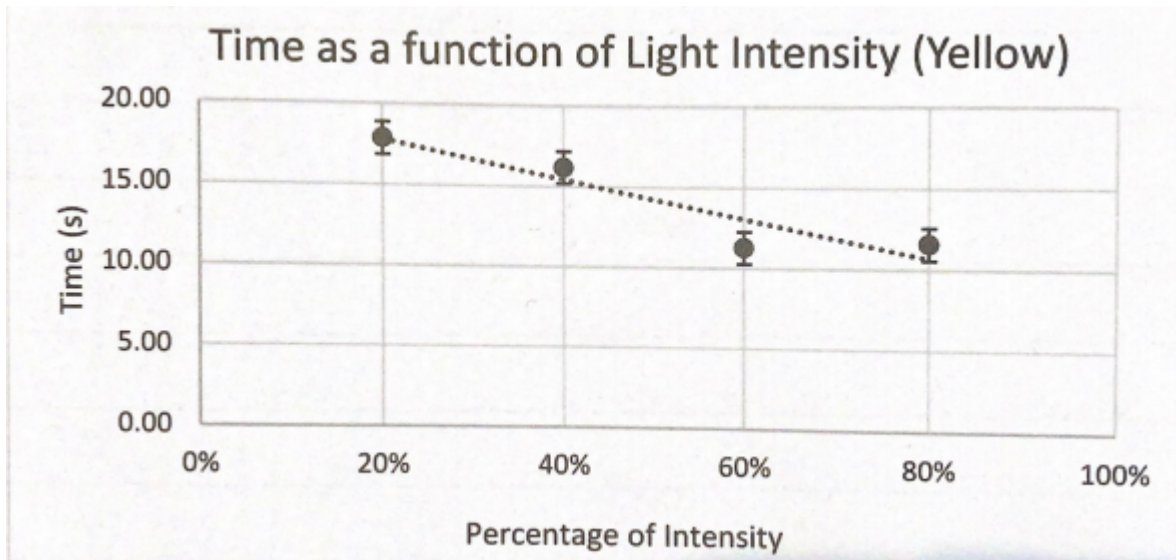


Figure 2: *graph of stabilisation time as a function of intensity (yellow light).*

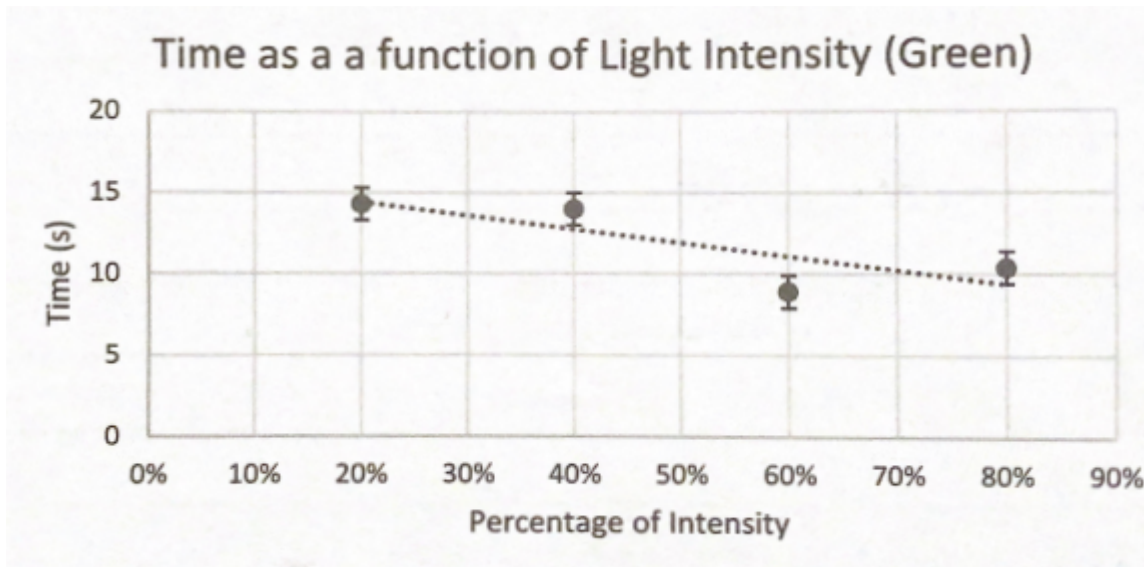


Figure 3: *graph of stabilisation time as a function of intensity (green light).*

However, we've also observed that the stopping potentials varied very little due to change in intensity ($\Delta V_s \approx 0.05V$). This is expected since the energy of emitted photoelectrons is only affected by the individual incident photons' energies, so they should not be affected by a change in the number of photons incident on the material since the energy of one photon will only have an effect on the energy of a single electron [3]. The small fluctuations of the stopping potentials could be attributed to light of a different frequency reaching the photodiode from other sources such as the lamps outside the booth where the experiment was conducted (the corridor and booths were only separated by a black curtain so light could've passed through in the gaps between the curtain and the walls).

When starting part (c), we found that as the photoelectric equation (equation (1)) predicts, the stopping potentials were directly proportional to the frequency of light. The graph for this can be seen on the next page.

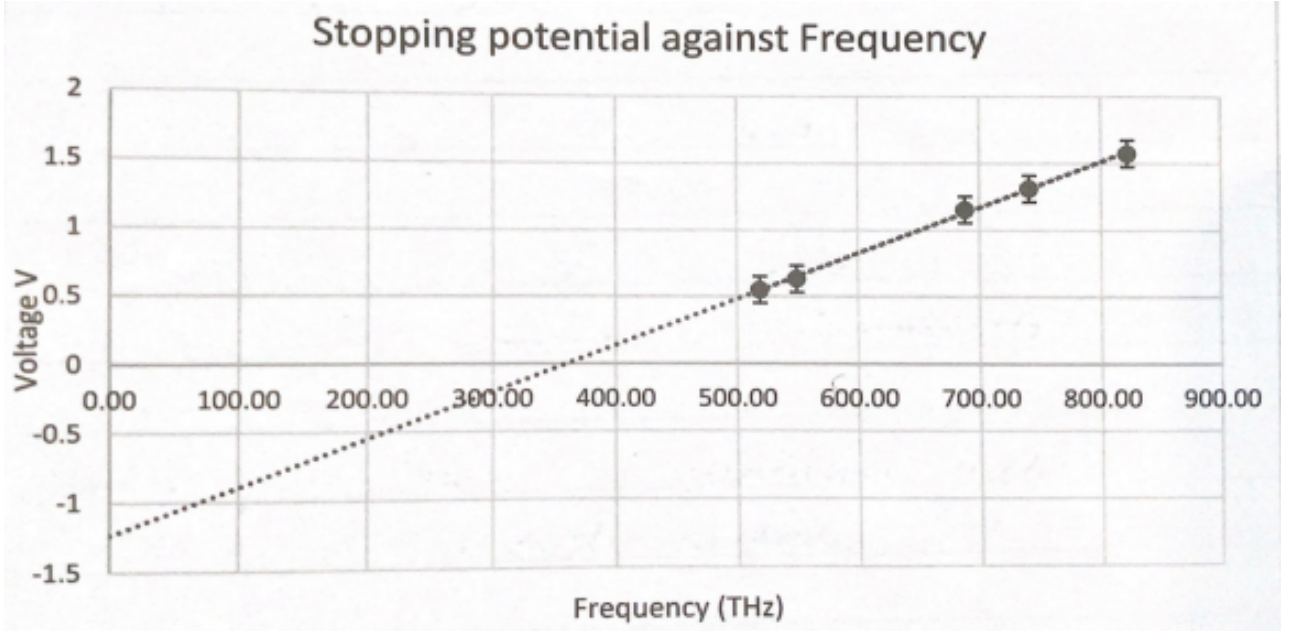


Figure 4: *graph of stopping potential against frequency.*

The gradient of the graph, m , was measured to be $(344 \pm 7.75) \cdot 10^{-17} VTHz^{-1}$. We find h by multiplying this value by e , the electron charge.

$$h = me \implies h = (344 \pm 7.75) \cdot 10^{-17} \cdot 1.6 \cdot 10^{-19} = (5.5 \pm 0.12) \cdot 10^{-34} Js \quad (5)$$

(Note: error in h is the error in the gradient multiplied by e . Error in m and the y -intercept were calculated using Excel.)

Reading off the y -intercept from the graph, we find the work function of the photodiode, ϕ , to be $(-1.24 \pm 0.05)eV$. So, finally we have our values for h and ϕ .

$$h = (5.5 \pm 0.12) \cdot 10^{-34} Js \quad (6)$$

$$\phi = (-1.24 \pm 0.05)eV \quad (7)$$

Discussion

In the end, we got a relatively accurate measurement of h with a maximum difference of 15.2% between our value and the literature value [5]. In the future, we could take multiple measurements of V_s for all frequencies of light investigated in order to reduce random error and obtain a more accurate result for m . Unfortunately, we couldn't find a literature value for ϕ as the photodiode manufacturer didn't list the material used.

The experiment confirmed the assumed relationship between V_s and f which can be seen in figure 4, yielding a precise value for the gradient and therefore h with only a 2.2% error.

Conclusion

Planck's constant and the work function of the photodiode have both been measured to a high degree of precision (2.2% error). It's clear on figure 4 that all 5 data points fit on a straight-line graph with very small error bars. This is most likely due to the high resolution of the multimeter used and the essentially negligible uncertainties in the wavelengths of the spectral

lines. However, unfortunately the literature value [5] doesn't fall within the error range. It could be possible to get a more accurate value for h by using a different element which has more spectral lines in the visible region (for example Iodine). This would give a larger data set and therefore reduce error.

It was also confirmed in this experiment that the intensity of light doesn't affect the photons' energies and hence the energies of the photoelectrons. Increasing the light intensity will simply increase the number of photoelectrons emitted as each photon will only emit a single photoelectron. This prediction can also be found in Einstein's original paper [3].

Acknowledgements

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References

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